

1B ANOVA

Assumptions:

Before conducting the ANOVA test, it is necessary to check whether the assumptions are satisfied. The first assumption is that the dependent variable, which is the plant (weight) should be continuous. In this data, plant (weight) is measured numerically, so this assumption is met. Another assumption is the independence observations. This means that the datapoint should not influence another. In this dataset, each plant was measured separately, so we can assume that the observations are independent. Other assumptions are listed down below.

Assumption #1: Dependent variable

The dependent variable in this data is the plant (weight), which is measured numerically at the continuous level (grams). So the assumption is satisfied

Assumption #2: Independent variable

The independent variable consists of three categorical groups: control (ctrl), treatment 1 (trt1), and treatment 2 (trt2).

Assumption #3: Independence

Based on the observations, there is no relationship between the two. Each plant was grown independently. So the assumption is met.

Assumption #4: Outliers

Boxplot was used to visually represent and show that there are no significant outliers across the three groups observed.

Assumption #5: Normality

All groups were approximately normally distributed. Each group and all the p-values were greater than 0.05 ($p > 0.05$), showing that the data is approximately normally distributed.

Assumption #6: Variances

The test was not significant, $F(2, 27) = 1.1$, $p = 0.341$, indicating variances are equal across groups.

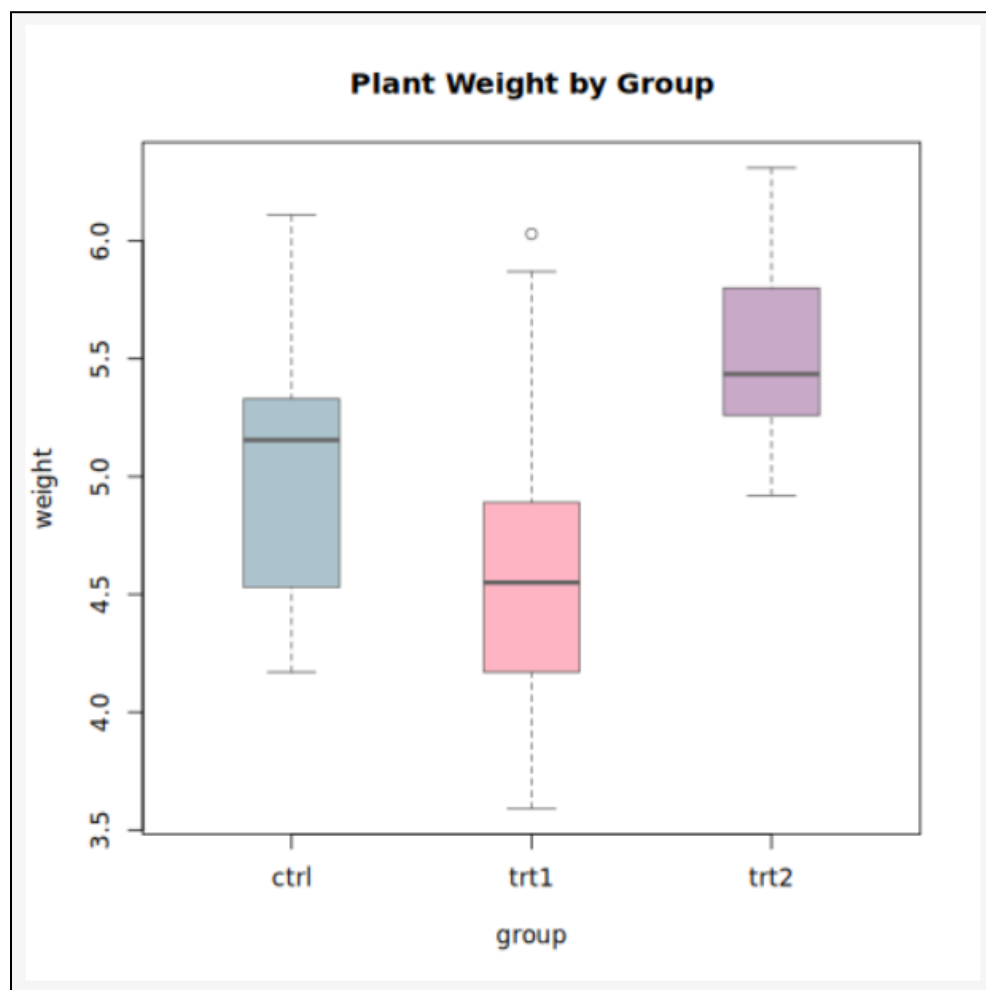
Null and Alternative Hypothesis

The goal of ANOVA test is to determine whether we have enough evidence to reject the null hypothesis. By comparing the computed p-value with the significance level, we can decide whether to reject or fail the null hypothesis. If the p-value is less than 0.05, then we reject the null hypothesis and conclude that there is a significant difference.

- **Null Hypothesis (H_0):** There is no significant difference in plant (weight) between the treatment groups.
- **Alternative Hypothesis (H_1):** There is a significant difference in plant (weight). And at least one group population is different from treatment groups.

Dataset and Problem

The PlantGrowth dataset consists of 30 observations divided equally among three groups: 10 plants in a control (ctrl), 10 in treatment group 1 (trt1), and 10 in treatment group 2 (trt2). Each group represents a different treatment condition applied to the plants. The goal is to determine whether these treatments result in different plant (weights), to test the effectiveness of different methods and conditions when applying to it.



Github:

<https://github.com/jensathensg/jens/blob/319191eaadf86dfe059994e57314ae74321c98/boxplot>

Descriptive Statistics of Plant Weight by Group

group	mean	median	sd.	min.	max.	n
control (ctrl)	5.0320000	5.1550000	0.5830914	4.1700000	6.1100000	10.0000000
treatment 1 (trt1)	4.6610000	4.5500000	0.7936757	3.5900000	6.0300000	10.0000000
treatment 2 (trt2)	5.5260000	5.4300000	0.4425733	4.9200000	6.3100000	10.0000000

Github:

<https://github.com/jensathensg/jens/blob/70888a68abc3ba77f01aabcdd4de9024d2569c7c/Descriptive%20Statistics>

Test of Homogeneity of variances

k-squared	df	p-value
2.8786	2	0.2371

ANOVA

	sum	sum of squares	Mean square	F	value
group	2	3.766	1.8832	4.846	0.0159
residuals	27	10.492	0.3886		

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Reporting:

The descriptive statistics of the PlantGrowth dataset show the summary of plant (weights) for the three groups including control (ctrl), treatment 1 (trt1), and treatment 2 (trt2). Each group has 100 observations, and the values include the mean, median, standard deviation, minimum, and maximum. The computed values help describe how the plants grew in each group.

Based on the observation and results of the statistics, the treatment 2 (trt2) group has the highest average weight with a mean of 5.526. This shows that treatment 2 may have helped the plants grow better compared to the other groups. On the other hand, the treatment 1 (trt1) group has the lowest mean of 4.661, which means the plants in this group did not grow as much as the two other groups. The control group is in the middle with a mean of 5.032.